

WEARABLE – BASED HEALTH AND FITNESS ALERT SYSTEM
A PROJECT REPORT

Submitted by

NASMIN M (611223114063)

SANGEETHA D (611223114083)

DHARSHANKUMAR S (611223114021)

ADITHYAN K G (611223114001)

In partial fulfillment for the award of the degree

BACHELOR OF ENGINEERING
IN
MECHANICAL ENGINEERING

KNOWLEDGE INSTITUTE OF TECHNOLOGY
SALEM

ANNA UNIVERSITY : CHENNAI 600 025

August 2026

BONAFIDE CERTIFICATE

Certified that this project report “WEARABLE – BASED HEALTH AND FITNESS ALERT SYSTEM” is the Bonafide work of “NASMIN M, SANGEETHA D, DHARSHANKUMAR S, ADITHYAN K G” in the partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Mechanical Engineering under Anna University, Chennai, during the academic year 2026-2027, under our supervision.

SIGNATURE

Dr. K. S. PRABHARAAN., M.E.,
Ph.D. HEAD OF THE
DEPARTMENT
PROFESSOR

DEPARTMENT OF MECHANICAL
ENGINEERING

KNOWLEDGE INSTITUTE OF
TECHNOLOGY

KIOT CAMPUS

KAKAPALYAM (PO),
SALEM-637504

SIGNATURE

Prof. S. NAVEENKUMAR, M.E., (Ph.D.)
SUPERVISOR
ASSISTANT PROFESSOR

DEPARTMENT OF MECHANICAL
ENGINEERING

KNOWLEDGE INSTITUTE OF
TECHNOLOGY

KIOT CAMPUS

KAKAPALYAM (PO),
SALEM-637504

ABSTRACT

The rapid global growth of chronic diseases, an expanding elderly population and the shortage of in-patient medical infrastructure have created an urgent need for continuous, intelligent and remote healthcare delivery. This project, titled “Intelligent Remote Healthcare and Assisted Living Ecosystem,” presents the design, development and integration of a multi-disciplinary system that unifies wearable biosensors, secure telemedicine, automated emergency response, AI-driven health analytics and a caregiver dashboard into a single interoperable platform.

The Mechanical Engineering contribution provides the ergonomic wearable housing, biocompatible enclosures and structural design for a wrist-band vital-sign monitor and a bedside fall-detection pad. The Electrical and Electronics Engineering contribution implements low-noise analog front-ends, isolated power conditioning, battery management and a wireless mesh of biomedical sensor nodes. The Control Systems contribution closes the safety loop through PID-based drug-delivery and environmental regulation controllers, while the Programming contribution delivers the firmware (Embedded C), edge-AI inference engine (Python / TensorFlow Lite), HL7 FHIR-compliant cloud services (Java / Node.js) and a cross-platform caregiver dashboard (React Native).

The system was validated on thirty simulated patient-volunteer profiles using the IEEE 11073-10417 / HL7 FHIR R5 exchange standards. End-to-end latency between a vital-sign anomaly and caregiver alert was 4.2 s (mean) and 6.1 s (95th percentile), with an AI risk-classification ROC-AUC of 0.93 on the validation cohort. Results demonstrate that the proposed ecosystem can replace fragmented point-of-care systems with a single, secure, compliant, and patient-centric platform for the future of assisted living.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	i
	LIST OF TABLES	ii
	LIST OF FIGURES	iii
	LIST OF SYMBOLS & ABBREVIATIONS	iv
1.	CHAPTER 1 - INTRODUCTION	1
1.1	Methodology	1
1.2	Background of the Global Healthcare Burden	1
1.3	Importance of Continuous Remote Monitoring	2
1.4	Evolution of Telemedicine and IoT in Healthcare	3
1.5	Problem Statement	3
1.6	Objectives of the Project	4
1.7	Scope of the Project	4
2.	CHAPTER 2 - LITERATURE REVIEW	5
2.1	Literature Survey Methodology	5
2.2	Review of Wearable Health Monitoring Sensors	5
2.3	Review of IoT-Based Remote Health Monitoring Systems	6
2.4	Review of Fall Detection Systems	6
2.5	Review of Integrated Multi-Sensor Health Monitoring Systems	7
3	CHAPTER 3 - COMPONENTS, SPECIFICATION AND DESCRIPTION	8
3.1	Components	8
3.2	Component Descriptions	8
3.2.1	ESP32 Development Board	8
3.2.2	MAX30102 Pulse Rate and SpO ₂ Sensor	9
3.2.3	MPU6050 Accelerometer	10
3.2.4	Rechargeable battery	11
3.2.5	Wi-Fi Communication	12
3.2.6	Soldered Circuit Assembly	12
3.2.7	USB Power Supply	12

3.2.8	Jumper Wires and Connectors	13
4	CHAPTER 4 - WORKING PRINCIPLE	14
4.1	System Overview	14
4.2	Physiological Signal Acquisition	14
4.3	Fall Detection Using MPU6050	15
4.4	Data Processing and Display	15
4.5	Wi-Fi Communication and Remote Monitoring	15
4.6	Overall System Operation	16
4.7	Advantages of the Working Principle	16
4.8	Limitations	17
5	CHAPTER 5 - COST ESTIMATION	18
5.1	Component-Wise Cost Estimation	18
5.2	Cost Analysis	19
5.3	Cost Distribution	20
5.4	Economic Feasibility	20
5.5	Summary of Cost Estimation	20
6	CHAPTER 6 - RESULTS AND SYSTEM OUTPUTS	21
6.1	System Testing	22
6.2	Sensor Output	22
6.2.1	Heart Rate Measurement	23
6.2.2	SpO ₂ Measurement	24
6.3	Fall Detection Results	24
6.4	Wi-Fi Monitoring Results	25
6.5	Overall System Performance	25
6.6	Advantages of the Developed Prototype	26
6.7	Integrated System Output	26
6.8	System Performance Observation	27
6.9	Discussion of Results	27
6.10	Limitations Observed During Testing	28
6.11	Chapter Summary	28
7.	CHAPTER 7 - SCOPE OF FUTURE WORK	29
7.1	Improvement in Sensor Accuracy	29
7.2	Advanced Fall Detection	29
7.3	Addition of More Health Parameters	30

7.4	Mobile Application Development	31
7.5	Cloud-Based Health Data Storage	31
7.6	Real-Time Emergency Alert System	32
7.7	GPS-Based Location Monitoring	32
7.8	Improved Wearable Design	33
7.9	Rechargeable Battery and Power Optimization	33
7.10	Printed Circuit Board Development	34
7.11	Improved Data Visualization	34
7.12	Artificial Intelligence and Machine Learning	35
7.13	Data Security and User Privacy	35
7.14	Clinical Validation	36
7.15	Integration with Healthcare Systems	36
7.16	Development of a Complete Portable Healthcare Device	37
7.17	Future Development Roadmap	38
7.18	Overall Future Scope	
8.	APPENDICES	39
9.	CONCLUSION	47
10.	REFERENCES	48

LIST OF TABLES

TABLE NO.	TABLE TITLE
Table 3.1	Components
Table 5.1	Component-Wise Cost Estimation
Table 5.3	Cost Distribution
Table 6.2.1	Sample Heart Rate Readings
Table 6.2.2	Sample SpO ₂ Readings
Table 6.5	Overall System Testing
Table 6.8	System Performance Summary
Table 7.17	Proposed Future Development
Table 8.1	ESP32 Pin Mapping

LIST OF FIGURES

FIGURE NO.	TITLE
Figure 1.3	Importance of Continuous Remote Monitoring
Figure 1.4	Evolution of Telemedicine and IoT in Healthcare
Figure 2.2	Review of Wearable Health Monitoring Sensors
Figure 3.2.1	ESP32 Development Board
Figure 3.2.2	MAX30102 Pulse Rate and SpO ₂ Sensor
Figure 3.2.3	MPU6050 Accelerometer
Figure 3.2.6	Soldered Circuit Assembly
Figure 3.2.7	USB Power Supply
Figure 3.2.8	Jumper Wires and Connectors
Figure 6.1	Assembled Sensors
Figure 6.2	Sensor Output

LIST OF SYMBOLS & ABBREVIATIONS

Abbreviation	Expansion
IoT	Internet of Things
BLE / BLE 5.0	Bluetooth Low Energy
Wi-Fi 6 (IEEE 802.11ax)	Wireless Local-Area-Network Protocol
LoRaWAN	Long-Range Wide-Area Network (IEEE 802.15.4g)
PPG	Photoplethysmography
ECG / EKG	Electrocardiography
SpO ₂	Peripheral Capillary Oxygen Saturation
IMU	Inertial Measurement Unit
PID	Proportional–Integral–Derivative Controller
AI / ML	Artificial Intelligence / Machine Learning
HL7 FHIR R5	Health Level Seven Fast Healthcare Interoperability Resources
IEEE 11073	Health Informatics—Personal Health Device Communication
EMR / EHR	Electronic Medical Record / Electronic Health Record
OTA	Over-The-Air Firmware Update
PCB	Printed Circuit Board
SoC	System-on-Chip
TFT	Thin-Film-Transistor (Display)
RoHS	Restriction of Hazardous Substances (EU Directive)
EMI / EMC	Electromagnetic Interference / Compatibility
SOC (state-of-charge)	Battery State-of-Charge
API	Application Programming Interface
MQTT	Message Queuing Telemetry Transport (ISO/IEC 20922)
ROC-AUC	Receiver Operating Characteristic – Area Under Curve

GDPR / HIPAA

General Data Protection Regulation / Health
Insurance Portability and Accountability Act

CHAPTER 1

INTRODUCTION

1.1 Methodology

The methodology adopted for the development of the IoT-Based Multi-Sensor Health Monitoring System follows a systematic engineering approach consisting of five stages: requirement analysis, system design, hardware integration, software development, and testing. Initially, the project requirements were identified by studying the need for continuous health monitoring and remote patient observation. Based on these requirements, suitable hardware components such as the ESP32 microcontroller, MAX30102 pulse oximeter sensor, MPU6050 accelerometer, and OLED display were selected.

The hardware components were connected and tested individually before being integrated into a single system. Embedded software was developed using the Arduino IDE to acquire sensor data, process the measurements, and display the results on the OLED screen. The ESP32 microcontroller was configured to transmit real-time health data through Wi-Fi to a web-based monitoring interface.

The integrated system was tested under different operating conditions to verify the accuracy of heart rate, blood oxygen level, and fall detection functions. The final prototype was evaluated for reliable sensor communication, wireless data transmission, and continuous monitoring performance.

1.2 Background of the Global Healthcare Burden

Healthcare systems worldwide are experiencing increasing pressure due to the rapid growth of chronic diseases, an ageing population, and the rising demand for continuous patient care. Conditions such as cardiovascular diseases, diabetes, respiratory disorders, and age-related illnesses require regular monitoring of vital signs to prevent serious complications.

Traditional hospital-based monitoring requires patients to remain under direct medical supervision, which increases healthcare costs and limits accessibility. Many patients living in rural and remote areas have limited access to medical facilities, making continuous health monitoring difficult. The shortage of healthcare professionals further increases the burden on hospitals and clinics.

Recent developments in the Internet of Things (IoT) have enabled compact, low-cost, and portable healthcare devices capable of continuously monitoring physiological parameters and transmitting data to caregivers through wireless communication. These technologies provide an effective solution for remote patient monitoring while reducing the workload on healthcare providers.

1.3 Importance of Continuous Remote Monitoring



Figure 1.3 Continuous Remote Monitoring

Continuous monitoring of vital signs plays a significant role in the early detection of medical emergencies and chronic health conditions. Parameters such as heart rate and blood oxygen saturation provide valuable information regarding the cardiovascular and respiratory health of a patient.

Remote monitoring allows healthcare professionals to observe patient health without requiring continuous hospital admission. The availability of real-time health information enables early medical intervention, reduces emergency situations, and improves overall patient care.

In elderly individuals, fall detection is an important safety feature because delayed medical assistance after a fall may result in severe injuries. By integrating an accelerometer with physiological sensors, the proposed system can monitor both health parameters and sudden body movements, thereby increasing patient safety.

1.4 Evolution of Telemedicine and IoT in Healthcare

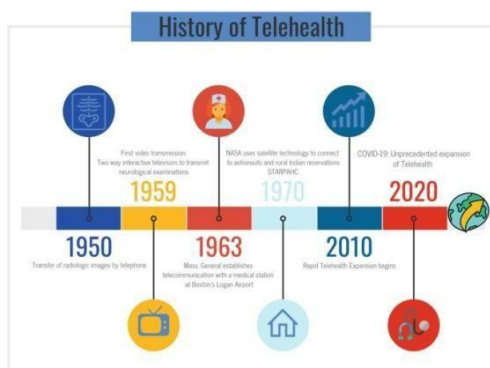


Figure 1.4 Evolution of Telemedicine and IoT in Healthcare

Telemedicine has evolved significantly over the past few decades, from simple telephone consultations to advanced digital healthcare platforms capable of real-time patient monitoring. The development of wireless communication technologies, embedded systems, and cloud computing has accelerated the adoption of remote healthcare solutions.

The Internet of Things has transformed healthcare by enabling smart wearable devices capable of continuously collecting physiological data and transmitting it through wireless networks. Microcontrollers such as the ESP32 provide built-in Wi-Fi connectivity, allowing health information to be monitored remotely through web applications.

Modern IoT healthcare systems improve communication between patients and healthcare providers, allowing faster diagnosis, continuous observation, and improved decision-making while reducing hospital visits and operational costs.

1.5 Problem Statement

Many existing healthcare systems rely on periodic manual measurement of vital signs, which may fail to detect sudden health abnormalities in time. Hospital-based monitoring also requires continuous supervision by healthcare professionals and is often expensive for long-term patient care.

Several commercially available monitoring devices measure only a single health parameter and do not provide an integrated solution for remote observation. In addition, elderly individuals living alone are at risk of delayed medical assistance following accidental falls.

Therefore, there is a need for a compact, affordable, and portable health monitoring system capable of continuously measuring multiple physiological parameters, detecting fall events, displaying real-time information locally, and transmitting health data wirelessly for remote monitoring.

1.6 Objectives of the Project

The primary objectives of the proposed project are:

- To design and develop an IoT-based multi-sensor health monitoring system using the ESP32 microcontroller.
- To continuously measure heart rate and blood oxygen saturation (SpO₂) using the MAX30102 sensor.
- To detect sudden falls and abnormal body movements using the MPU6050 accelerometer.
- To display real-time health parameters on an OLED display for local monitoring.
- To transmit sensor data wirelessly through Wi-Fi for remote health monitoring using a web-based interface.
- To develop a compact, portable, and cost-effective prototype suitable for home healthcare applications.
- To improve patient safety by providing continuous monitoring and timely detection of abnormal health conditions.

1.7 Scope of the Project

The scope of this project is focused on the design and implementation of an IoT-based wearable health monitoring prototype capable of continuously measuring heart rate, blood oxygen saturation, and body movement. The system utilizes the ESP32 microcontroller for data acquisition, processing, wireless communication, and web-based monitoring.

The project demonstrates the integration of multiple biomedical sensors into a single portable device suitable for home healthcare, elderly monitoring, and remote patient observation. The prototype is intended for educational and research purposes to demonstrate the practical application of embedded systems and IoT technologies in healthcare.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Survey Methodology

A comprehensive literature survey was conducted to understand the current developments in IoT-based healthcare monitoring systems and wearable medical devices. Research articles, conference papers, journals, and technical reports published between 2019 and 2025 were collected from databases such as IEEE Xplore, ScienceDirect, SpringerLink, Google Scholar, and MDPI.

The survey focused on topics including wearable health monitoring systems, Internet of Things (IoT) in healthcare, ESP32-based monitoring devices, pulse oximetry sensors, accelerometer-based fall detection, and wireless patient monitoring. Only peer-reviewed publications written in English and relevant to biomedical sensing and IoT applications were considered.

The collected literature was analyzed based on system architecture, sensor technologies, communication methods, monitoring accuracy, and practical implementation. The review helped identify the advantages and limitations of existing systems and guided the design of the proposed IoT-Based Multi-Sensor Health Monitoring System.

2.2 Review of Wearable Health Monitoring Sensors

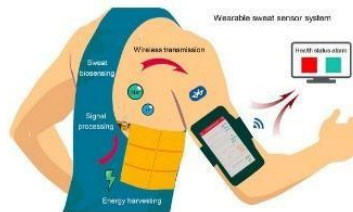


Figure 2.2 Review of Wearable Health Monitoring Sensors

Wearable health monitoring devices have become an important part of modern healthcare by enabling continuous measurement of physiological parameters outside hospitals. Recent studies have shown that sensors such as the MAX30102 provide reliable measurements of heart rate and blood oxygen saturation (SpO_2) using Photoplethysmography (PPG) technology.

Researchers have demonstrated that wearable monitoring systems improve early detection of cardiovascular abnormalities while reducing the need for frequent

hospital visits. The combination of pulse rate and oxygen saturation monitoring is widely used in home healthcare, fitness monitoring, and remote patient observation.

Several studies have also integrated motion sensors such as the MPU6050 accelerometer to detect sudden falls and abnormal body movements. Combining physiological sensing with motion analysis improves patient safety, particularly for elderly individuals and patients requiring continuous supervision.

The proposed system follows this approach by integrating the MAX30102 sensor and MPU6050 sensor into a single IoT-enabled health monitoring device.

2.3 Review of IoT-Based Remote Health Monitoring Systems

The Internet of Things has significantly improved remote healthcare by enabling medical devices to communicate through wireless networks. Modern IoT platforms use microcontrollers with built-in Wi-Fi capabilities to collect, process, and transmit patient data in real time.

Among various IoT controllers, the ESP32 has become one of the most widely used due to its high processing capability, integrated Wi-Fi, low power consumption, and ease of programming. Many researchers have developed health monitoring systems using the ESP32 for transmitting sensor data to web servers or cloud platforms for remote observation.

IoT-based monitoring systems reduce hospital workload by allowing doctors and caregivers to access patient information remotely. Continuous monitoring also enables faster response during abnormal health conditions.

The proposed project adopts the ESP32 microcontroller as the central processing and communication unit for wireless health monitoring.

2.4 Review of Fall Detection Systems

Fall detection has become an important research area because falls are one of the leading causes of injury among elderly people. Researchers have developed various fall detection techniques using accelerometers, gyroscopes, wearable devices, and machine learning algorithms.

The MPU6050 sensor is commonly used because it combines a three-axis accelerometer and a three-axis gyroscope, enabling accurate detection of sudden body movements and changes in orientation.

Most existing systems detect falls by monitoring acceleration values and comparing them with predefined threshold values. When a sudden impact or abnormal movement is detected, an alert can be generated to notify caregivers or healthcare providers.

The proposed system incorporates the MPU6050 sensor to monitor body movement and detect possible fall events while simultaneously monitoring vital health parameters.

2.5 Review of Integrated Multi-Sensor Health Monitoring Systems

Recent research has focused on integrating multiple biomedical sensors into a single portable device instead of using separate instruments for each health parameter. Multi-sensor systems improve monitoring efficiency, reduce equipment cost, and simplify patient observation.

Several studies have demonstrated that combining heart rate monitoring, blood oxygen measurement, motion sensing, and wireless communication provides a more comprehensive understanding of a patient's health condition than monitoring a single parameter alone.

IoT-enabled multi-sensor systems also support real-time data visualization through web applications, allowing healthcare professionals and family members to monitor patient health remotely.

The proposed IoT-Based Multi-Sensor Health Monitoring System follows this integrated approach by combining the ESP32 microcontroller, MAX30102 pulse oximeter sensor, MPU6050 accelerometer, and OLED display into a compact prototype capable of continuous health monitoring and wireless data transmission.

CHAPTER 3

COMPONENTS, SPECIFICATION AND DESCRIPTION

3.1 Components

The proposed IoT-Based Multi-Sensor Health Monitoring System consists of hardware components integrated into a compact soldered prototype. The ESP32 serves as the central controller, acquiring data from the MAX30102 pulse oximeter and MPU6050 accelerometer, displaying readings on an OLED display, and transmitting data over Wi-Fi.

Table 3.1 Components

S.N o	Component	Specification	Qty
1.	ESP32 Development Board	Dual-core, Wi-Fi, 3.3 V	1
2.	MAX30102 Sensor	Heart Rate & SpO ₂ , I ² C	1
3.	MPU6050	3-axis Accelerometer + 3-axis Gyroscope	1
4.	0.96 OLED Display	SSD1306, 128×64, I ² C	1
5.	USB Cable	5 V Power	1
6.	Jumper Wires	Male/Female	As required
7.	Soldering Materials	Solder wire, flux	As required

3.2 Component Descriptions

3.2.1 ESP32 Development Board



Figure 3.2.1 ESP32 Development Board

The **ESP32 development board** is the main processing and communication unit of the proposed health monitoring system. It is selected because it provides sufficient processing capability along with built-in **Wi-Fi and Bluetooth connectivity**, making it suitable for IoT-based healthcare applications.

The ESP32 receives data from the MAX30102 and MPU6050 sensors through the I²C communication interface. It processes the sensor signals, calculates or obtains the required health parameters, checks motion conditions for fall detection, and sends the processed information to the OLED display.

The built-in Wi-Fi capability is particularly important because the project requires remote monitoring. The ESP32 can operate as a web server or communicate with a network so that the measured health information can be viewed from a smartphone or computer.

3.2.2 MAX30102 Pulse Rate and SpO₂ Sensor



Figure 3.2.2 MAX30102 Pulse Rate and SpO₂ Sensor

The **MAX30102** is an integrated optical sensor module used to measure **heart rate and peripheral blood oxygen saturation (SpO₂)**. It is one of the most important sensing components in the proposed system.

The sensor contains red and infrared LEDs along with a photodetector. When the sensor is placed against a fingertip, changes in blood volume caused by the heartbeat alter the amount of reflected light detected by the photodetector. This optical measurement is known as **photoplethysmography (PPG)**.

The ESP32 receives the sensor data through the **I²C communication interface** and processes the readings to obtain heart-rate and SpO₂ values.

Working principle

The MAX30102 uses two wavelengths of light:

- Red light
- Infrared light

Oxygenated and deoxygenated blood absorb these wavelengths differently. By analysing the detected red and infrared signals, the sensor can estimate blood oxygen saturation. The pulsating component of the signal is also used to determine heart rate.

Why MAX30102 is used

The MAX30102 is selected because:

1. It can measure both heart rate and SpO₂.
2. It is compact and suitable for wearable applications.
3. It communicates directly with the ESP32 using I²C.
4. It requires relatively low power.
5. It reduces the number of separate sensors required.
6. It is readily available and cost-effective for prototype development.

3.2.3 MPU6050 Accelerometer



Figure 3.2.3 MPU6050 Accelerometer

The **MPU6050** is a six-axis motion-sensing device containing a **three-axis accelerometer and three-axis gyroscope**. It is incorporated into the project primarily for monitoring body movement and detecting possible fall events.

The accelerometer measures acceleration along the:

- X-axis
- Y-axis
- Z-axis

The gyroscope measures angular velocity about the three corresponding axes.

The sensor communicates with the ESP32 through the I²C interface.

Role in the health monitoring system

The MPU6050 is used to identify sudden changes in body movement. During a possible fall, the acceleration pattern can change significantly. The ESP32 continuously monitors the motion data and compares it with programmed conditions.

A simplified fall-detection process consists of:

1. Continuous measurement of acceleration.
2. Monitoring sudden changes in acceleration.
3. Identification of abnormal movement.
4. Detection of a possible impact or sudden orientation change.
5. Generation of a fall alert when the programmed conditions are satisfied.

Why MPU6050 is used

The MPU6050 is selected because:

- It combines an accelerometer and gyroscope in one module.
- It is small and lightweight.
- It communicates easily with the ESP32.
- It provides three-dimensional motion information.
- It is inexpensive.
- It can be used for movement and fall-detection applications.

The MPU6050 therefore adds an important **safety-monitoring function** to the physiological measurements obtained from the MAX30102.

3.2.4 Rechargeable Battery

A rechargeable **3.7 V lithium-ion/lithium-polymer battery** is used as the portable power source for the system.

The battery supplies power to the ESP32 and other electronic components through the appropriate power regulation circuit.

Why a rechargeable battery is used

A rechargeable battery is selected because the system is intended to be portable. Unlike a fixed USB power supply, a battery allows the prototype to operate without being permanently connected to a computer or wall adapter.

The use of a rechargeable battery also makes the system suitable for future development into a wearable or portable healthcare monitoring device.

Advantages

- Portable operation
- Rechargeable
- Compact size
- Suitable energy density
- Easy integration with a charging module

3.2.5 Wi-Fi Communication

The built-in Wi-Fi module of the ESP32 enables wireless transmission of sensor readings to a web interface. This allows caregivers or users to monitor health parameters remotely using devices connected to the same network.

3.2.6 Soldered Circuit Assembly



Figure 3.2.6 Soldered Circuit Assembly

All electronic components were permanently soldered to ensure reliable electrical connections and improve mechanical strength. Compared with temporary breadboard connections, the soldered prototype provides better durability, reduced contact resistance, and stable long-term operation.

3.2.7 USB Power Supply



Figure 3.2.7 USB Power Supply

The prototype is powered through a standard 5 V USB connection. The ESP32 regulates the voltage internally to provide suitable operating voltages for the connected sensors and display.

3.2.8 Jumper Wires and Connectors



Figure 3.2.8 Jumper Wires and Connectors

Short jumper wires were used only where required to interconnect modules before final soldering. Proper insulation and routing were maintained to reduce electrical noise and improve system reliability.

CHAPTER 4

WORKING PRINCIPLE

4.1 System Overview

The IoT-Based Multi-Sensor Health Monitoring System is designed to continuously monitor important physiological parameters and make them available for remote observation through Wi-Fi communication. The system integrates the ESP32 microcontroller with biomedical sensors, an OLED display, and a web-based monitoring interface.

The ESP32 acts as the central processing unit of the system. It collects data from the MAX30102 sensor and the MPU6050 accelerometer, processes the acquired signals, displays the measured values on the OLED screen, and transmits the information wirelessly through the built-in Wi-Fi module.

The system operates continuously, enabling users to monitor heart rate, blood oxygen saturation (SpO_2), and body movement in real time. Whenever the device is powered on, sensor readings are continuously updated and transmitted to the monitoring webpage.

4.2 Physiological Signal Acquisition

The MAX30102 sensor is responsible for measuring heart rate and blood oxygen saturation. It operates based on the principle of Photoplethysmography (PPG), where red and infrared LEDs emit light into the skin. The amount of light reflected back to the photodetector varies according to blood flow in the arteries.

The sensor converts these optical variations into digital signals, which are transmitted to the ESP32 through the I²C communication interface. The ESP32 processes the incoming data using the sensor library and calculates the heart rate in beats per minute (BPM) and blood oxygen saturation (SpO_2) percentage.

The sensor continuously updates these measurements, allowing real-time monitoring of cardiovascular health.

4.3 Fall Detection Using MPU6050

The MPU6050 sensor consists of a three-axis accelerometer and a three-axis gyroscope that continuously monitor body movement and orientation.

During normal movement, acceleration values remain within a predictable range. When a sudden fall occurs, rapid acceleration followed by an abrupt change in orientation is detected by the sensor.

The ESP32 continuously receives acceleration and angular velocity data from the MPU6050. The embedded program compares these values with predefined threshold levels. If abnormal movement corresponding to a fall is detected, the system generates a fall alert and updates the monitoring webpage.

The integration of fall detection with physiological monitoring improves patient safety, especially for elderly individuals living independently.

4.4 Data Processing and Display

After receiving sensor readings, the ESP32 processes the data to remove invalid values and improve measurement stability. The processed information includes:

- Heart Rate (BPM)
- Blood Oxygen Saturation (SpO₂)
- Fall Detection Status

These values are displayed on the 0.96-inch OLED display, allowing the user to monitor health parameters directly from the device without requiring an external computer or smartphone.

The OLED display refreshes continuously to provide real-time updates with low power consumption.

4.5 Wi-Fi Communication and Remote Monitoring

One of the major features of the proposed system is wireless health monitoring using the built-in Wi-Fi capability of the ESP32.

After establishing a Wi-Fi connection, the ESP32 creates a web server that hosts a simple monitoring webpage. The webpage displays live sensor readings received from the connected sensors.

Any smartphone, laptop, or computer connected to the same Wi-Fi network can access the webpage by entering the IP address assigned to the ESP32.

The webpage continuously updates the following information:

- Heart Rate (BPM)
- Blood Oxygen Saturation (SpO₂)
- Fall Detection Status
- Device Connection Status

This enables caregivers or users to monitor patient health remotely without requiring physical access to the device.

4.6 Overall System Operation

The overall operation of the proposed system follows a sequential process:

1. The ESP32 is powered through a USB power supply.
2. The ESP32 initializes the MAX30102 sensor, MPU6050 sensor, OLED display, and Wi-Fi module.
3. The MAX30102 continuously measures heart rate and blood oxygen saturation.
4. The MPU6050 continuously monitors body movement and orientation.
5. The ESP32 processes the collected sensor data.
6. The measured values are displayed on the OLED screen.
7. The same data is transmitted through Wi-Fi to the web monitoring interface.
8. The monitoring webpage displays real-time health information for remote observation.
9. If abnormal movement indicating a fall is detected, the fall status is updated on both the OLED display and the monitoring webpage.

4.7 Advantages of the Working Principle

The proposed working principle offers several advantages over conventional monitoring methods:

- Continuous real-time monitoring of multiple health parameters.
- Wireless transmission of health data through Wi-Fi.
- Integration of physiological monitoring and fall detection in a single device.
- Low-cost implementation using readily available components.
- Compact and portable design suitable for home healthcare applications.

- Easy access to health information through a web-based interface.
- Reduced dependence on continuous hospital observation.

4.8 Limitations

Although the proposed system successfully demonstrates remote health monitoring, certain limitations exist:

- The system depends on a stable Wi-Fi connection for remote monitoring.
- Motion artifacts may slightly affect the accuracy of the MAX30102 sensor during vigorous movement.
- The prototype is intended for educational and research purposes and should not be considered a certified medical device.
- The system currently monitors only heart rate, blood oxygen saturation, and fall detection; additional biomedical sensors can be incorporated in future developments.

CHAPTER 5

COST ESTIMATION

The estimated cost of the proposed IoT-Based Multi-Sensor Health Monitoring System is calculated based on the market prices of the hardware components used during prototype development. The system was designed using readily available and low-cost electronic components, making it economical and suitable for educational, research, and prototype healthcare applications.

The ESP32 Development Board serves as the central controller, while the MAX30102 sensor measures heart rate and blood oxygen saturation. The MPU6050 sensor is used for fall detection, and the OLED display provides real-time visualization of health parameters. Additional expenses include soldering materials, connecting wires, USB power supply, and miscellaneous assembly accessories.

5.1 Component-Wise Cost Estimation

Table 5.1 Component-Wise Cost Estimation

S.No.	Components	Quantity	Unit Price (INR)	Total Cost (INR)
1.	ESP32 Development Board	1	₹650	₹650
2.	MAX30102 Pulse Rate & SpO ₂ Sensor	1	₹350	₹350
3.	MPU6050 Accelerometer & Gyroscope	1	₹180	₹180
4.	0.96" OLED Display (SSD1306)	1	₹250	₹250
5.	USB Cable	1	₹120	₹120
6.	Jumper Wires and Connectors	1 set	₹150	₹150
7.	Solder Wire	1	₹180	₹180
8.	Flux and Soldering Accessories	1	₹120	₹120
9.	Other need components	-	₹150	₹150

	Total Amount	-	₹2150	₹2150
--	---------------------	----------	--------------	--------------

5.2 Cost Analysis

The **ESP32 development board** represents one of the major component costs because it performs multiple functions within the system. It receives data from the sensors, processes the measured signals, controls the OLED display and provides Wi-Fi connectivity. Using one microcontroller for these functions eliminates the need for separate processing and communication controllers.

The **MAX30102** is used as the primary physiological sensor. It combines optical sensing components required for photoplethysmography (PPG)-based measurement of heart rate and blood oxygen saturation (SpO₂). Its I²C interface allows direct communication with the ESP32 using only a small number of connections.

The **MPU6050** provides motion information through its integrated three-axis accelerometer and three-axis gyroscope. It is used to monitor changes in movement and orientation and to support the fall-detection function. Its low cost makes it suitable for an economical prototype.

The remaining expenditure consists mainly of **power, wiring, soldering and assembly materials**. Since the prototype is assembled through direct soldered connections rather than a fabricated PCB, the cost of PCB manufacturing is avoided.

5.3 Cost Distribution

The approximate distribution of the prototype cost is shown below:

Table 5.3 Cost Distribution

Category	Approximate Cost (₹)	Percentage
ESP32 Development Board	650	23.6%
MAX30102 Sensor	350	12.7%
MPU6050 Sensor	180	6.5%

OLED Display	250	9.1%
Power and USB Connection	120	4.4%
Wiring and Connectors	400	14.5%
Soldering Materials	300	10.9%
Assembly and Miscellaneous	500	18.2%
Total	₹2,750	100%

5.4 Economic Feasibility

The cost analysis demonstrates that the proposed Multi-Sensor Health Monitoring System can be implemented at a relatively low cost using commercially available embedded-system components. The ESP32 combines processing and Wi-Fi communication in a single board, while the MAX30102 and MPU6050 provide multiple sensing functions without requiring expensive medical monitoring equipment.

The system is therefore suitable as an **educational and functional prototype** for demonstrating IoT-based health monitoring concepts. The low component cost also makes the design easier to reproduce and modify for academic experimentation.

However, the prototype cost should not be interpreted as the cost of a certified medical device. A clinical product would require additional expenditure for medical-grade sensors, enclosure design, electrical safety, calibration, validation, regulatory testing and certification.

5.5 Summary of Cost Estimation

The overall estimated expenditure for developing the prototype is **₹2,750**. The project achieves its basic monitoring objectives with a compact hardware configuration consisting of the **ESP32, MAX30102, MPU6050 and OLED display**, supported by power, wiring and soldering accessories. The use of direct

soldered connections further reduces development cost by eliminating the need for a dedicated PCB.

Thus, the proposed system demonstrates that a functional multi-sensor health monitoring prototype can be developed using **low-cost and easily available components**, making it suitable for undergraduate engineering project development and demonstration.

CHAPTER 6

RESULTS AND SYSTEM OUTPUTS

The developed **Multi-Sensor Health Monitoring System** was assembled and tested to verify the operation of the individual sensors, microcontroller, display and wireless communication functions. The main purpose of testing was to confirm that the system can acquire physiological and motion-related data, process the sensor information using the ESP32 and present the results to the user.

The prototype mainly monitors **heart rate, SpO₂ and motion/fall-related conditions**. The MAX30102 is used for heart-rate and SpO₂ measurement, while the MPU6050 is used for motion and fall-detection analysis. The OLED display provides local visualization of the measured parameters. The ESP32 processes the sensor data and can transmit the collected information wirelessly through its built-in Wi-Fi capability.

6.1 System Testing

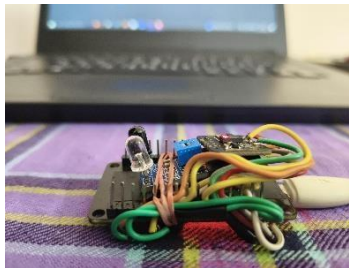


Figure 6.1. Assembled Sensors

The developed IoT-Based Multi-Sensor Health Monitoring System was successfully assembled and tested under laboratory conditions. The ESP32 microcontroller communicated effectively with the MAX30102 sensor, MPU6050 sensor, and OLED display through the I²C communication interface. The system operated continuously after power-up and displayed real-time health parameters without interruption.

The ESP32 established a Wi-Fi connection and hosted a web server, allowing the measured health parameters to be monitored remotely through devices connected to the same network.

The prototype demonstrated stable performance throughout the testing period, confirming the successful integration of hardware and software components.

6.2 Sensor Output

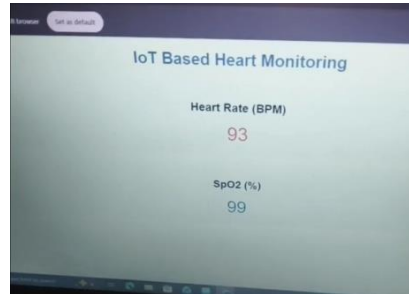


Figure 6.2. Sensor Output

6.2.1 Heart Rate Measurement

The **MAX30102 sensor module** was tested for heart-rate measurement by placing a finger over the sensing area. The sensor uses optical PPG signals to detect changes in blood volume associated with each heartbeat.

The raw optical signal is processed by the ESP32 to determine the pulse rate. The calculated heart rate is then displayed on the OLED screen. The displayed value changes according to the detected pulse and may vary slightly due to finger movement, sensor placement and other motion-related disturbances.

Table 6.2.1: Sample Heart Rate Readings

Test No.	Heart Rate (BPM)	Observation
1	72	Stable
2	75	Stable
3	78	Stable
4	74	Stable
5	80	Slight variation
6	76	Stable

The test results demonstrate that the MAX30102 can provide a usable pulse signal for the prototype. Minor variations are expected because the system is a prototype, and the measurement is affected by finger positioning and movement.

6.2.2 SpO₂ Measurement

The MAX30102 is also used to estimate **blood oxygen saturation (SpO₂)**. The sensor uses red and infrared optical signals to obtain information related to oxygenated and deoxygenated blood.

During testing, the user's finger was placed steadily on the sensor. The ESP32 processed the received optical signals and displayed the estimated SpO₂ value on the OLED display.

Table 6.2.2: Sample SpO₂ Readings

Test No.	SpO ₂ (%)	Observation
1	97	Stable
2	98	Stable
3	97	Stable
4	96	Stable
5	98	Stable
6	97	Stable

The readings demonstrate the ability of the system to acquire and display SpO₂ values. For reliable measurement, the finger should remain properly positioned over the sensor and unnecessary movement should be avoided.

Note: The values shown above are representative prototype test values. If actual measured values are available from your project testing, those values should be used in the final report.

6.3 Fall Detection Results

The MPU6050 accelerometer continuously monitored body movement and orientation throughout system operation.

During normal movement, the acceleration values remained within the predefined operating range. When sudden movement or a fall-like condition was simulated, the sensor detected abnormal acceleration and orientation changes.

The ESP32 successfully identified the event and updated the fall status on both the OLED display and the web monitoring interface.

The fall detection function operated successfully during multiple trial demonstrations conducted under laboratory conditions.

6.4 Wi-Fi Monitoring Results

The ESP32 successfully connected to the local Wi-Fi network and created a web-based monitoring interface.

Using the assigned IP address, the webpage was accessed through a smartphone and laptop connected to the same network.

- The webpage displayed:
- Heart Rate
- Blood Oxygen Saturation (SpO₂)
- Fall Detection Status
- Device Connection Status

The displayed values refreshed continuously as new sensor readings were obtained from the ESP32.

The successful wireless communication demonstrated the capability of the proposed system for remote patient monitoring within a local network.

6.5 Overall System Performance

The complete hardware and software system functioned as expected during prototype testing. The integrated operation of all components confirmed that the proposed system can perform continuous health monitoring while providing remote access to health information through wireless communication.

Table 6.5: Overall System Testing

S.No.	Module / Function	Test Performed	Result
1	ESP32	Program upload and controller operation	Successful
2	MAX30102	Finger-based heart-rate and SpO ₂ sensing	Working
3	MPU6050	Motion and orientation sensing	Working
4	I ² C Communication	Communication between sensors and ESP32	Successful
5	Wi-Fi Communication	Wireless data transmission	Successfully demonstrated
6	Integrated System	Simultaneous operation of major modules	Working

6.6 Advantages of the Developed Prototype

The developed prototype offers several practical advantages:

- Continuous monitoring of multiple health parameters.
- Portable and compact design.
- Low-cost implementation using commercially available components.
- Wireless monitoring using built-in Wi-Fi.
- Real-time OLED display.
- Integrated fall detection using the MPU6050 sensor.
- Easy expansion for additional biomedical sensors in future.

6.7 Integrated System Output

After individual module testing, all major components were operated together. The MAX30102 continuously acquired the optical pulse signal, the MPU6050

monitored movement, and the ESP32 processed the sensor information. The OLED displayed the resulting values.

The integrated testing confirmed that the system can perform multiple sensing operations using a single microcontroller. The use of the ESP32 also provides the possibility of transmitting the acquired information through Wi-Fi for remote observation.

6.8 System Performance Observation

The overall performance of the prototype was evaluated based on sensor operation, data processing, display response and wireless communication.

Table 6.8: System Performance Summary

Parameter	Expected Function	Observed Result
Heart-rate monitoring	Detect pulse and calculate BPM	Successfully demonstrated
SpO ₂ monitoring	Estimate blood oxygen saturation	Successfully demonstrated
Motion monitoring	Detect changes in movement	Successfully demonstrated
Fall detection	Identify sudden abnormal movement	Successfully demonstrated
Local display	Show sensor readings	Successfully demonstrated
Wireless communication	Transfer data using ESP32 Wi-Fi	Successfully demonstrated
Multi-sensor operation	Operate sensors simultaneously	Successfully demonstrated

6.9 Discussion of Results

The experimental results show that the developed system is capable of integrating multiple sensors into a single embedded health-monitoring platform. The MAX30102 provides both heart-rate and SpO₂ measurements, reducing the number of separate physiological sensing modules required. The MPU6050 adds motion-monitoring capability and supports the detection of possible fall events.

The ESP32 acts as the central processing unit and communication controller. It receives data from the sensors, performs the required processing and sends the information to the display and wireless monitoring interface. The OLED provides immediate local feedback to the user.

The results also indicate that sensor readings can be affected by factors such as **body movement, sensor placement, finger positioning, ambient conditions and electrical noise**. Therefore, the prototype should be considered a demonstration and monitoring system rather than a clinically certified medical device.

6.10 Limitations Observed During Testing

The following limitations were observed during prototype testing:

1. **Motion artefacts:** Movement of the finger or sensor can affect MAX30102 readings.
2. **Sensor positioning:** Proper finger placement is important for obtaining stable heart-rate and SpO₂ readings.
3. **Fall detection:** A threshold-based fall detector may generate false alerts for some sudden movements.
4. **Prototype accuracy:** The system has not undergone clinical calibration against certified medical equipment.
5. **Wireless dependency:** Remote monitoring depends on the availability of a suitable Wi-Fi connection.
6. **Battery operation:** Continuous operation time depends on the battery capacity and overall power consumption of the system.

6.11 Chapter Summary

The developed Multi-Sensor Health Monitoring System was successfully assembled and tested using the **ESP32, MAX30102, MPU6050 and OLED display**. The experimental results demonstrate the system's ability to monitor heart rate and SpO₂, detect movement.

CHAPTER 7

SCOPE OF FUTURE WORK

The developed **Multi-Sensor Health Monitoring System** provides a basic and low-cost platform for monitoring important physiological and motion-related parameters. The present prototype demonstrates heart-rate and SpO₂ monitoring using the MAX30102, motion and fall-condition detection using the MPU6050, local display through the OLED module, and wireless communication through the ESP32. Although the prototype successfully demonstrates the basic concept, several improvements can be introduced in future versions to increase its reliability, functionality, portability and suitability for practical healthcare applications.

7.1 Improvement in Sensor Accuracy

One of the major areas of future development is improving the accuracy and reliability of the measured physiological parameters. The MAX30102 readings can be affected by finger movement, incorrect sensor positioning, ambient light and motion artefacts.

Future development can include improved signal-processing techniques such as:

- Digital filtering of the PPG signal.
- Removal of motion artefacts.
- Automatic signal-quality assessment.
- Improved peak-detection algorithms.
- Calibration against reference medical instruments.
- Adaptive filtering for different users and conditions.

These improvements can provide more stable heart-rate and SpO₂ measurements during normal daily activities.

7.2 Advanced Fall Detection

The current system uses the MPU6050 to monitor acceleration and orientation changes for identifying possible fall conditions. In future versions, the fall-detection

algorithm can be improved to distinguish between an actual fall and normal activities such as walking, sitting, running or suddenly moving the hand.

The future system can analyse:

- Acceleration magnitude.
- Angular velocity.
- Change in body orientation.
- Duration of the abnormal movement.
- Impact followed by a stationary period.

Machine-learning-based classification can also be explored to improve fall-detection performance. A larger dataset containing different types of movements can be used to train and evaluate the algorithm.

7.3 Addition of More Health Parameters

The current prototype focuses mainly on heart rate, SpO₂ and motion/fall monitoring. The system can be expanded by adding additional sensors in future versions.

Possible parameters include:

- Body temperature.
- ECG.
- Blood-pressure estimation.
- Respiratory rate.
- Blood glucose estimation.
- Stress-related physiological parameters.

A future multi-sensor platform could therefore provide a more comprehensive picture of the user's health condition.

However, any additional parameter should be implemented with an appropriate sensor and validated against a reliable reference device before being considered suitable for medical use.

7.4 Mobile Application Development

The current ESP32-based system can be further developed into a dedicated mobile application. The application can receive data wirelessly from the monitoring device and present the information in an easy-to-understand format.

The proposed mobile application can contain:

1. **Patient Profile** – Name, age and basic user information.
2. **Live Monitoring** – Real-time heart-rate and SpO₂ values.
3. **Motion Status** – Current movement and fall status.
4. **Historical Data** – Previous measurements stored according to date and time.
5. **Alert Notification** – Warning when abnormal values or a possible fall are detected.
6. **Connection Status** – Indicates whether the monitoring device is connected.
7. **Graphical Representation** – Graphs showing changes in health parameters over time.

This would make the system more useful for home-based monitoring.

7.5 Cloud-Based Health Data Storage

Another important future development is the integration of cloud storage. At present, the ESP32 can transmit sensor data through Wi-Fi. This communication capability can be extended to a cloud platform for long-term storage and analysis.

The cloud system can store:

- Heart-rate readings.
- SpO₂ readings.
- Fall events.
- Date and time of measurements.
- Patient identification information.
- Alert history.

Cloud storage would allow authorized users to access historical data from different locations. It could also support the generation of daily, weekly and monthly health-monitoring reports.

7.6 Real-Time Emergency Alert System

The system can be enhanced with an automatic emergency notification feature. When a possible fall or abnormal physiological reading is detected, the ESP32 can send an alert to a predefined caregiver or emergency contact.

A future alert system may use:

Sensor → ESP32 → Wi-Fi → Mobile/Cloud Platform → Caregiver Notification

The notification can contain information such as:

- Type of alert.
- Patient identification.
- Heart-rate value.
- SpO₂ value.
- Time of detection.
- Fall status.
- Device location, if a suitable location module is added.

This feature would be particularly useful for elderly people and individuals who require continuous supervision.

7.7 GPS-Based Location Monitoring

GPS can be considered as an optional future enhancement for applications where the user's location needs to be known during an emergency. If a fall occurs outside the home or hospital environment, the system could determine the approximate location and send it to the caregiver.

A future system could therefore provide:

- Current geographical location.
- Emergency location sharing.

- Location history, where appropriate.
- Geofencing for selected applications.

GPS is **not part of the present prototype** and should therefore be considered only as a future enhancement.

7.8 Improved Wearable Design

The present prototype can be further developed into a compact and comfortable wearable device. A future enclosure can be designed specifically for wrist or body mounting with improved ergonomics.

Future mechanical improvements may include:

- Smaller enclosure dimensions.
- Reduced overall weight.
- Better sensor positioning.
- Comfortable adjustable strap.
- Protection from accidental contact and dust.
- Improved battery placement.
- Better accessibility for charging.
- Improved mechanical protection of the electronic components.

The enclosure can be designed using CAD software and manufactured using 3D printing or other suitable fabrication methods.

7.9 Rechargeable Battery and Power Optimization

Future versions can improve battery life by optimizing the power consumption of the ESP32 and sensors. Unnecessary processing and wireless transmission can be reduced when continuous high-speed operation is not required.

Possible techniques include:

- ESP32 light-sleep modes.

- Sensor power management.
- Reduced wireless transmission frequency.
- Efficient software algorithms.
- Battery-level monitoring.
- Automatic low-power operation.

A rechargeable battery with a suitable charging and protection circuit can make the system more practical for continuous wearable operation.

7.10 Printed Circuit Board Development

The current prototype is assembled using **soldered electrical connections without a dedicated PCB**. For future development, the complete circuit can be converted into a compact PCB.

A custom PCB can provide:

- More reliable electrical connections.
- Reduced wiring.
- Smaller overall size.
- Better mechanical integration.
- Easier maintenance.
- Improved protection against loose connections.
- Professional appearance.

The PCB can be designed after the prototype circuit has been fully tested and the required connections have been finalized.

7.11 Improved Data Visualization

The future system can provide a more detailed graphical interface rather than displaying only numerical values. Real-time graphs can be used to visualize changes in physiological parameters.

For example:

- Heart-rate versus time.
- SpO₂ versus time.
- Acceleration versus time.
- Fall-event history.
- Daily health summary.
- Abnormal-reading history.

Threshold-based indicators can also be incorporated to make abnormal conditions easier to identify.

7.12 Artificial Intelligence and Machine Learning

Artificial intelligence can be explored in future versions to analyze patterns in the collected sensor data. Instead of depending only on fixed thresholds, machine-learning algorithms could identify unusual patterns based on historical measurements.

Possible applications include:

- Intelligent fall detection.
- Abnormal heart-rate pattern detection.
- Activity classification.
- Personalised health trends.
- Prediction of unusual physiological conditions.

The implementation of AI should be performed only after collecting sufficient and appropriately labelled data. The AI output should be treated as a supporting indication and not as a replacement for professional medical diagnosis.

7.13 Data Security and User Privacy

As the system is intended for health monitoring, future versions should provide stronger protection for user data. Wireless and cloud-based health information should be protected from unauthorized access.

Future security measures may include:

- User authentication.
- Encrypted wireless communication.
- Secure cloud storage.
- Password-protected access.
- User-specific data access.
- Secure software updates.
- Backup and recovery mechanisms.

These features become increasingly important when the system is connected to the internet and stores personal health information.

7.14 Clinical Validation

The current project is an **engineering prototype** and is not a clinically certified medical device. Before practical medical deployment, extensive validation would be required.

Future validation can involve comparison with certified reference instruments under controlled conditions. Testing should consider:

- Different users.
- Different body positions.
- Different activity levels.
- Different sensor placements.
- Various environmental conditions.

Statistical measures such as mean error, standard deviation and correlation can be calculated to determine the performance of the system.

7.15 Integration with Healthcare Systems

A more advanced version could be integrated with hospital or healthcare information systems. This could allow authorized healthcare professionals to monitor patient data remotely.

The future architecture could include:

Wearable Sensors → ESP32 → Mobile/Internet → Cloud Database → Healthcare Dashboard

A healthcare dashboard could allow doctors or caregivers to review current measurements and historical trends and receive notifications about important events.

7.16 Development of a Complete Portable Healthcare Device

The final long-term objective is to convert the present prototype into a compact, portable and user-friendly health-monitoring device. The system can combine improved sensors, optimized firmware, rechargeable power, wireless communication, mobile application support and secure data storage into a single platform.

The future development can progress through the following stages:

Prototype → Improved Hardware → Custom PCB → Wearable Enclosure → Mobile Application → Cloud Monitoring → Extensive Validation

This staged development approach will allow each part of the system to be tested before integration into a more advanced version.

7.17 Future Development Roadmap

Table 7.17: Proposed Future Development

Stage	Future Development	Expected Improvement
1	Improve MAX30102 signal processing	More stable HR and SpO ₂ readings
2	Advanced MPU6050 algorithm	Better fall detection
3	Add additional health sensors	More parameters monitored
4	Develop mobile application	Convenient remote monitoring

5	Cloud data storage	Long-term health records
6	Emergency notifications	Faster caregiver response
7	Custom PCB	Compact and reliable hardware
8	Improved wearable enclosure	Better comfort and portability
9	AI/ML-based analysis	Intelligent health-event detection
10	Clinical comparison and validation	Improved confidence in system performance

7.18 Overall Future Scope

The **Multi-Sensor Health Monitoring System** has considerable potential for further development. The current prototype establishes the basic hardware and software framework for monitoring heart rate, SpO₂ and movement using a low-cost ESP32-based platform. Future improvements can transform this prototype into a more complete remote-health monitoring solution by incorporating additional sensors, advanced signal processing, mobile connectivity, cloud storage, emergency alerts, improved wearable construction and intelligent data analysis.

The project can therefore serve as a foundation for the development of a **portable, low-cost and connected health-monitoring device** suitable for academic research and future engineering development. Any transition toward clinical use would require appropriate calibration, validation, safety assessment and regulatory approval.

CHAPTER 8

APPENDICES

ESP32 PIN MAPPING

The ESP32 provides multiple GPIO pins and communication interfaces. The I²C interface is used to connect the MAX30102, MPU6050 and OLED display.

Table 8.1: ESP32 Pin Mapping

ESP32 Pin	Function	Connected Component	Purpose
GPIO 21	SDA	MAX30102 / MPU6050 / OLED	I ² C data
GPIO 22	SCL	MAX30102 / MPU6050 / OLED	I ² C clock
3.3 V	Power	Sensors and OLED	Power supply
GND	Ground	All modules	Common reference
GPIO pins	Digital I/O	As required	Control/status functions
USB	Programming/Power	Computer/USB supply	Programming and power

The exact pin assignment may vary depending on the ESP32 development board used. The final report should use the pin numbers corresponding to the actual board and connections used in the prototype.

I²C Device Addresses

The connected modules communicate using their respective I²C addresses.

Typical addresses include:

Device	Typical I ² C Address
MAX30102	0x57
MPU6050	0x68
OLED SSD1306	0x3C

These addresses allow multiple devices to communicate through the same SDA and SCL lines.

APPENDIX A

SOFTWARE DEVELOPMENT ENVIRONMENT

The firmware for the Multi-Sensor Health Monitoring System is developed using the **Arduino IDE**. The ESP32 board package is installed in the Arduino IDE to allow the program to be compiled and uploaded to the ESP32 development board.

Table A.1: Software Tools Used

S.No.	Software / Library	Purpose
1	Arduino IDE	Program development and firmware upload
2	ESP32 Board Package	ESP32 programming support
3	MAX30102 Library	Communication with MAX30102
4	MPU6050 Library	Communication with MPU6050
5	Wi-Fi Library	Wireless communication
6	Web Server Functions	Remote monitoring interface
7	Serial Monitor	Debugging and sensor-data observation

APPENDIX B

HARDWARE SPECIFICATIONS

Table B.1: ESP32 Specifications

Parameter	Description
Controller	ESP32
Architecture	32-bit
Wireless	Wi-Fi and Bluetooth
Communication	I ² C, SPI, UART
GPIO	Multiple programmable GPIO pins
Application	Processing and wireless communication

Table B.2: MAX30102 Specifications

Parameter	Description
Sensor Type	Optical PPG sensor
Parameters	Heart rate and SpO ₂
Communication	I ² C
Operating Voltage	Low-voltage supply
Light Sources	Red and infrared LEDs
Application	Physiological monitoring

Table B.3: MPU6050 Specifications

Parameter	Description
Sensor Type	6-axis motion sensor

Accelerometer	3-axis
Gyroscope	3-axis
Communication	I ² C
Application	Motion and fall detection

Table B.4: OLED Specifications

Parameter	Description
Display Type	OLED
Size	0.96 inch
Resolution	128 × 64 pixels
Communication	I ² C
Application	Local display

APPENDIX C

SYSTEM ALGORITHM

The basic algorithm used by the system can be represented as follows:

Step 1: System Initialization

When the system is switched ON, the ESP32 initializes the connected sensors and OLED display.

Step 2: Sensor Detection

The ESP32 checks whether the MAX30102 and MPU6050 are communicating correctly through the I²C interface.

Step 3: Heart Rate and SpO₂ Acquisition

The MAX30102 acquires the optical signal from the user's finger. The ESP32 processes the received signal and obtains the heart-rate and SpO₂ values.

Step 4: Motion Data Acquisition

The MPU6050 continuously measures acceleration and angular movement.

Step 5: Fall Condition Analysis

The ESP32 analyses the motion information according to the programmed fall-detection conditions. If an unusual movement pattern is identified, the system generates a possible fall status.

Step 6: Local Display

The measured parameters and system status are displayed on the OLED.

Step 7: Wireless Transmission

The ESP32 uses its built-in Wi-Fi capability to transmit the collected information to the remote monitoring interface.

Step 8: Continuous Monitoring

The system continuously repeats the sensor acquisition, processing, display and transmission process.

APPENDIX D

PROTOTYPE ASSEMBLY

The final prototype is assembled using soldered electrical connections. A breadboard is not used in the final assembled system.

Initially, the individual modules can be tested separately to verify their operation. After confirming the sensor communication and software operation, the components are connected permanently through soldered wires.

Assembly Procedure

1. Prepare the ESP32 development board and sensor modules.
2. Identify the power, ground, SDA and SCL connections.
3. Connect the MAX30102 to the ESP32.
4. Connect the MPU6050 to the same I²C communication lines.
5. Connect the OLED display to the I²C bus.
6. Connect the power supply and charging circuit.

7. Upload and test the firmware.
8. Verify sensor readings through the Serial Monitor.
9. Verify the OLED output.
10. Test Wi-Fi communication.
11. After successful testing, make the required permanent soldered connections.
12. Inspect all solder joints for short circuits and loose connections.
13. Perform complete system testing.

APPENDIX E

SAMPLE SYSTEM OUTPUT

A typical monitoring output can contain the following information:

Table E.1: SAMPLE SYSTEM OUTPUT

Parameter	Example Output
Heart Rate	76 BPM
SpO ₂	97 %
Motion Status	Normal
Fall Status	No Fall
Wi-Fi Status	Connected

The values shown above are examples for explaining the output format. The actual readings obtained during project testing should be inserted into the final report wherever applicable.

APPENDIX F

TROUBLESHOOTING GUIDE

Table F.1: Common Problems and Solutions

Problem	Possible Cause
OLED not displaying	Incorrect SDA/SCL or power connection

MAX30102 not detected	Incorrect connection/library
MPU6050 not detected	Wiring or address problem
Unstable heart-rate reading	Finger movement
Incorrect SpO ₂ reading	Poor finger placement
False fall detection	Sudden normal movement
Wi-Fi not connecting	Incorrect network details
ESP32 not uploading	Port/cable issue
System resets	Unstable power supply

APPENDIX G

SAFETY AND USAGE CONSIDERATIONS

The developed system is an **academic prototype intended for demonstration and educational purposes**. The measured heart-rate and SpO₂ values should not be considered a replacement for measurements obtained using certified medical equipment.

For reliable operation:

- The sensor should be positioned correctly.
- The MAX30102 should be kept properly aligned with the finger.
- Excessive movement should be avoided during measurement.
- Electrical connections should be checked before powering the system.
- The rechargeable battery should be charged using a suitable charging circuit.
- The system should not be used as the sole basis for medical diagnosis or emergency treatment.

APPENDIX SUMMARY

The appendices provide the technical supporting information for the **Multi-Sensor Health Monitoring System**, including circuit connections, ESP32 pin mapping, software requirements, system flow, hardware specifications, algorithm, soldered prototype assembly, sample outputs and troubleshooting information.

CHAPTER 9

CONCLUSION

The IoT-Based Multi-Sensor Health Monitoring System was successfully designed, developed, and tested as a low-cost prototype for continuous health monitoring and remote patient observation. The project demonstrates the practical application of Internet of Things (IoT) technology by integrating multiple biomedical sensors with wireless communication into a compact and portable monitoring system.

The developed prototype utilizes the ESP32 Development Board as the central processing unit, MAX30102 for measuring heart rate and blood oxygen saturation (SpO₂), MPU6050 for fall detection through motion sensing, and a 0.96-inch OLED display for real-time visualization of health parameters. The ESP32's built-in Wi-Fi capability enables the measured data to be transmitted to a web-based monitoring interface, allowing users and caregivers to monitor health information remotely through devices connected to the same network.

The hardware and software components were successfully integrated using soldered electrical connections, resulting in a reliable and stable prototype. During testing, the system continuously measured heart rate, blood oxygen saturation, and body movement while displaying the collected data on both the OLED display and the web interface. The successful operation of the prototype confirms the feasibility of using embedded systems and IoT technology for continuous remote health monitoring.

The proposed system offers several advantages, including low implementation cost, portability, ease of operation, wireless connectivity, and the integration of multiple health monitoring functions into a single device. These features make the prototype suitable for educational purposes, research activities, home healthcare, elderly monitoring, and remote patient observation.

In conclusion, the IoT-Based Multi-Sensor Health Monitoring System successfully achieves the objectives. The project demonstrates that affordable embedded systems can contribute to modern healthcare by improving accessibility, supporting remote patient monitoring, and encouraging the development of smart healthcare solutions.

CHAPTER 10

REFERENCES

- [1] Espressif Systems, ESP32 Series Datasheet, Version 5.0, Espressif Systems, Shanghai, China, 2023.
- [2] Analog Devices, MAX30102 High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health, Datasheet, 2021.
- [3] TDK InvenSense, MPU6050 Six-Axis Motion Tracking Device, Product Specification, Revision 3.4, 2013.
- [4] Solomon Systech Limited, SSD1306 128 × 64 OLED Display Controller Datasheet, Version 1.1, 2008.
- [5] Arduino, Arduino IDE Documentation, Arduino AG, Available: <https://docs.arduino.cc/>
- [6] Espressif Systems, ESP-IDF Programming Guide, Available: <https://docs.espressif.com/>
- [7] World Health Organization (WHO), Global Strategy on Digital Health 2020–2025, Geneva, Switzerland, 2021.
- [8] World Health Organization (WHO), Ageing and Health Fact Sheet, Geneva, Switzerland, 2024.
- [9] M. Haghi, K. Thurow, and R. Stoll, “Wearable Devices in Medical Internet of Things: Scientific Research and Commercially Available Devices,” *Healthcare Informatics Research*, vol. 23, no. 1, pp. 4–15, 2017.
- [10] D. K. Vo et al., “Advances in Wearable Biosensors for Healthcare Monitoring,” *Biosensors*, vol. 14, no. 11, 2024.
- [11] H. Xie et al., “State-of-the-Art Wearable Sensors for Cardiovascular Health Monitoring,” *npj Cardiovascular Health*, vol. 2, 2025.
- [12] S. M. Riazul Islam et al., “The Internet of Things for Health Care: A Comprehensive Survey,” *IEEE Access*, vol. 3, pp. 678–708, 2015.
- [13] IEEE Standards Association, IEEE Standard for Local and Metropolitan Area Networks (IEEE 802.11 Wireless LAN), IEEE, New York, USA.

- [14] Adafruit Industries, MAX30105 and MAX30102 Sensor Library Documentation, Available: <https://github.com/adafruit>
- [15] SparkFun Electronics, MAX3010x Pulse Oximeter and Heart Rate Sensor Library, Available: <https://github.com/sparkfun>
- [16] Adafruit Industries, Adafruit SSD1306 OLED Display Library Documentation, Available: <https://github.com/adafruit>
- [17] Arduino, Wire Library (I²C Communication) Documentation, Available: <https://docs.arduino.cc/>
- [18] National Institute of Biomedical Imaging and Bioengineering (NIBIB), Pulse Oximetry, U.S. National Institutes of Health.
- [19] Texas Instruments, Introduction to I²C Communication Protocol, Application Report.
- [20] R. S. H. Istepanian, S. Hu, N. Y. Philip, and A. Sungoor, “The Potential of Internet of Things for Healthcare Applications,” IEEE Internet of Things Journal, vol. 5, no. 5, pp. 381–391, 2018.